

Project Design Phase-II Data Flow Diagram & User Stories

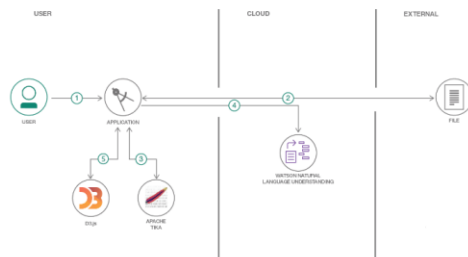
Date	01 October 2026
Team ID	SWTID-2026-2391
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) for the Script-Controlled ACL project shows how users and administrators interact with the ServiceNow interface, how the ACL Evaluation Engine checks roles and the READ script, and how permitted CRUD operations are applied to the Institution Details table (u_institution_details). The project uses ServiceNow authentication and authorization; the main data and access-control flow is described in the supplied project documentation.

DFD Level 0 – Context Diagram

Flow



1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.

Example: DFD Level 0 (Industry Standard)

User Stories

Use the below template to list all the user stories for the product.

DFD Level 0

User / Administrator	ServiceNow Application (ACL-controlled access)	Institution Details Records
Login / Record request CRUD request	Authenticate → Evaluate roles, branch condition and READ script → Permit / Deny	u_institution_details (Student, Faculty, Branch, Email, Phone, Description)
Access result / permitted UI actions	READ / CREATE / WRITE / DELETE decision	Records shown or modified

Flow: User/Administrator → ServiceNow Interface → ACL Evaluation Engine → Role/Branch/Script checks → CRUD decision → Institution Details records → permitted result. Administrators receive full administrative access as documented in the project.

DFD Level 1 – ACL-Controlled Record Access

Step	Process	Input / Check	Output
1	User authentication	ServiceNow login	Authenticated session
2	Access request	Open list/form or request CRUD operation	Requested operation
3	ACL evaluation	Operation: READ / CREATE / WRITE / DELETE	ACL evaluation started
4	Role / condition / script check	bb1, bb2, bb3, bb4; branch value; READ script	Allow or deny
5	Record access	u_institution_details	View / create / edit / delete if allowed
6	Verification	Impersonated user or administrator	Expected access result

Data Store: u_institution_details contains Student Roll Number, Student Name, Faculty Name, Branch (ECE/EEE/CSE), Email, Phone Number and Description. The READ ACL is advanced and uses a server-side script; CREATE, WRITE and DELETE are controlled by the corresponding roles.

User Stories

The following user stories are derived from the supplied Script-Controlled ACL project documentation. They describe the demonstrated access-control functions rather than adding unsupported application features.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
EEE User	Authentication / Access	USN-1	As an EEE User, I can log in to the ServiceNow instance so that I can access the Institution Details application.	I can authenticate through the ServiceNow instance and access the permitted application interface.	High	Sprint-1
EEE User	READ / Record Visibility	USN-2	As an EEE User with role bb1, I can view the permitted EEE records in the Institution Details table.	The Student Records list displays the intended EEE records and restricts unauthorized record access.	High	Sprint-1
User without required role	Authorization / Restricted Access	USN-3	As a user without the required access role, I cannot view records protected by the READ ACL.	The READ ACL denies access when the required role condition is not satisfied.	High	Sprint-1
EEE User	CREATE	USN-4	As an authorized user with role bb2, I can create a new Institution Details record.	The New/create capability is available and a new record can be created when bb2 is assigned.	High	Sprint-1
EEE User	WRITE	USN-5	As an authorized user with role bb3, I can edit an Institution Details record.	The edit/write capability is available when bb3 is assigned.	High	Sprint-1
EEE User	DELETE	USN-6	As an authorized user with role bb4, I can delete an Institution Details record.	The delete capability is available when bb4 is assigned.	High	Sprint-1
EEE User	Combined CRUD Authorization	USN-7	As a user with bb1, bb2, bb3 and bb4, I can view, create, edit and delete records according to the configured ACLs.	READ, CREATE, WRITE and DELETE capabilities are available as specified by the role assignments.	High	Sprint-2
Administrator	Administrative Access	USN-8	As an administrator, I can access the Institution Details records with full administrative access.	The administrator can view and perform the documented CRUD operations on the records.	High	Sprint-1
Administrator	ACL Configuration	USN-9	As an administrator/security administrator, I can configure the READ, CREATE, WRITE and DELETE ACLs.	The ACL records can be created or modified after obtaining the required security privileges.	High	Sprint-1
Tester	Verification / Impersonation	USN-10	As a tester, I can impersonate users with different role	The Student Records list/form shows the expected	Medium	Sprint-2

			combinations and verify the expected access.	permissions for each tested role combination.		
Administrator	Branch Classification	USN-11	As an administrator, I can create Institution Details records with ECE, EEE or CSE branch values.	The Branch field accepts the documented choices ECE, EEE and CSE.	Medium	Sprint-1
Administrator	Access Validation	USN-12	As an administrator, I can verify that incorrect role assignments or ACL changes produce the expected access behavior.	The relevant user is impersonated and the access result is checked after configuration changes.	Medium	Sprint-2

Source basis: Script-Controlled ACL – Restrict Record Access Based on Field Value project documentation supplied with this task. The project documentation identifies ServiceNow as the platform, u_institution_details as the custom table, bb1–bb4 as the custom roles, and separate READ/CREATE/WRITE/DELETE ACL controls.